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Original Article

Isolation and molecular identification of extended spectrum beta-lactamase producing bacteria from urinary tract infection

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ABSTRACT

Background: Recent years, the treatment of multi-drug resistant bacteria and their effect are very difficult due to the virulence factors modification. Based on the world wide thread, we have tried to identify the ESBLs producing bacteria from urinary tract infection patients. In result, the highly antibiotic resistant effect of *Pseudomonas aeruginosa* and *Klebsiella pneumoniae* were identified.

Methods: Initially, Hexa disc diffusion method was performed to detect the multi-drug resistant bacteria using respective antibiotics of HX066, HX033 and HX077, HX012 discs. Consecutively, the ESBL producing ability of confirmed multi drug resistant bacteria was performed to detect their ESBL producing ability using specific extended spectrum beta lactamase (ESBLs) detection discs of Hexa G-minus 24. Furthermore, the ESBL producing ability of the bacteria was confirmed by ESBLs detection Ezy MIC^{TM} E-test stripe method.

Results and conclusions: In result, 10, 5 and 4 mm and 10, 14 and 8 mm zone of inhibition against imipenem (IPM), Ticarcillin/Clavulanic acid (TCC), Cefoperazone (CPZ) and Ampicillin (AMP), Norfloxacin (NX), Nalidixic acid (NA) antibiotics for *P. aeruginosa* and 16, 22 and 10, 18 mm zone of inhibition against ceftazidime (CAZ), methicillin (MET), ampicillin amoxyclav (AMC), co-trimoxazole (COT) of the HX077 HX012 for *K. pneumoniae* were observed. Based on the Clinical & Laboratory Standards Institute (CLSI) guidelines, both the bacteria were more resistant to tested antibiotics and it could be developed more resistant against all the tested antibiotics. In addition, the phenotypic detection of ESBL production effect was also performed against both the selected uropathogens, and the results were shown ≥22 mm, ≥27 zone of inhibition against all the tested antibiotics. Further, the genetic identification of multi plux PCR result was shown TEM, SHV and CTX-m genes were present in both the selected uropathogens. Finally, our results were correlated each other and concluded that the selected uropathogens were multi drug resistant effect and also ESBLs producer.

Introduction

Urinary tract infections (UTIs) are the most prevalent bacterial infections that presumably cause one-half of all people during their lifetime due to the extensive use of β-lactam antibiotics [1]. Resistance of pathogenic organisms to countenance antibiotics is a clinical challenge all over the world [2]. In particular, the development of antibiotics against gram negative bacteria (GNB) faces more difficult challenges than Gram positive bacteria (GPP) due to the production of ESBLs. It is a heterogenous group of plasmid mediated bacterial enzyme, that confer the ability of hydrolyzing and inactivating a wide variety of beta-lactam antibiotics and penicillin derivatives (penams), cephalosporins (cephems), monobactams, and carbapenems [3]. It also exhibits co-resistance to other classes of antibiotics including fluoroquinolones, trimethoprim-sulfamethoxazole and aminoglycosides, resulting in the limitation

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of treatment option. Due to these reasons, the ESBLs producing bacterial infection reports have been increased worldwide [4].

The main reason for production of ESBLs in bacteria is development of multi drug resistant (MDR) effect. Among the microbes, bacteria are the most predominant microbes to produce resilience behavioral character through variety of mechanisms. Enabling of gram negative bacteria and their characteristic nature is more suitable to develop resistant effect against many antibiotic families through genetic mechanisms. The antibiotics alone or combination with any other antibiotics are produced co-resistant enzymes and it find out very difficulties such as decreased membrane permeability, dissemination, expression of over production of efflux pump and continuous modification of penicillin-binding proteins. Commonly, MDR bacteria are a worldwide thread and known to increase high morbidity and mortality in health care. Prevalence of infection by MDR bacteria cause severe infection and prolonged stay in human body [5].

The first types of ESBLs are the derivatives of temoniera (TEM-1), sulfhydryl variable (SHV-1) enzymes, reported in the early 1980s due to the point mutation mainly in *K. pneumonia* strains associated with nosocomial outbreaks and highly prevalence of emerging CTX-M enzymes were reported in late 1980s with different epidemiology and it highly found in most of the UTI causing GNB [6,7]. Recently, most of the Enterobacteriaceae and *Pseudomonas* sp. becoming a emerging health problem due to the ESBLs production including *Escherichia coli*, *Proteus mirabilis*, *Klebsiella pnumoniae*, *Enterobacter* sp., *Acinetobacter baumannii* and *Pesudomonas aeruginosa* [8]. The production of ESBLs from Enterobacteriaceae are vary in prevalence and are becoming increasingly treatment failure in clinical practice worldwide. They developed resistance to third-generation antibiotics such as cefotaxime, ceftriaxone, ceftazidime and oxyimino-monobactams over a period [9]. Some combination therapies are currently available to inhibit the ESBLs; they are Augmentin (amoxicillin/clavulanate), Zosyn (piperacillin/tazobactam), and recently Avycaz (ceftazidime/avibactam) [10]. However, to date no any equivalent therapy is available to inhibit the ESBLs producing Gram negative bacteria, so far no anti-ESBL drug has been recommended clinically by WHO and currently used anti-ESBL drugs are not effective [11,12]. There are only a few reports are evidently reported regarding clinical efficacy of therapeutics in treatment of infections caused by ESBL producing GNB. Still, it is an emerging challenge to discover anti-ESBL inhibitor, particularly broad-spectrum ESBL inhibitors, given the structural diversity and mechanistic complexities of ESBLs.

Materials and methods

Detection of multi-drug resistant profile

For detection of multi-drug resistant (MDR) effect, 100 numbers of *P. aeruginosa* and *K. pnumoniae* (each one of 50 strains), which cause UTIs were procured from K.A.P.V. Government Medical College & Hospital, Tiruchirappalli, Tamil Nadu, India. The MDR patterns of selected uropathogens detected by disc diffusion method using specific UTI panel of *P. aeruginosa* and *K. pnumoniae* discs such as HX066, HX033 and HX077, HX012 respectively. All the tested antibiotics were followed by the CLSI guidelines (2016), including cefotaxime (CTX-10 μg), imipenom (IPM-10 μg), Ticarcillin/clavulanic acid (TCC 75/10 μg), Cefepime (CPM 30 μg), Ceftriaxone (CTR 30 μg) Cefoperazone (CPZ 75 μg), Ampicillin (AMP-10 μg), Co-Trimoxazole (25 μg), Norfloxacin (NX-10 μg), nalidixic acid (NA-30 μg), Netillin (NET-300 μg), cephalothin (CEP-30 μg), for *P. aeruginosa* and amoxyclav (AMC-30 μg), ampicillin (AMP-10 μg), ciprofloxacin (Cip-5 μg), Co-Trimoxazole (Cot-25 μg) gentamicin (Gen-10 μg), norfloxacin (NX10 μg) & ciprofloxacin (CIP-5 μg), imipenem (IPM-10 μg), meropenem (MRP-10 μg), ertapenem (ETP-10 μg), cefoperazone/sulbactum (CFS-75/30 μg) piperacillin/tazobactam (PIT-100/10 μg) for *Klebsiella pnumoniae* as proposed method of Rajivgandhi et al. [13].

Screening of ESBL producers

The ESBL producing ability of selected MDR uropathogens *P. aeruginosa* and *Klebsiella pneumoniae* were determined by phenotypic disc diffusion assay using Hexa G-minus 24 disc method (CLSI guidelines 2016) including cefpodoxime (CPD -10 μg), cefpodoxime/clavulanic acid (CCL – 10/5 μg), ceftazidime (CAZ 30 μg), ceftazidime/clavulanic acid (CAC – 30/10 μg), cefotaxime (CTX – 30 μg), cefotaxime/clavulanic acid (CEC – 30/10 μg). For confirmation of ESBLs production, the antibiotic discs CAZ and CTX were exhibited as $\geq$ 24 and $\geq$27, whereas $\geq$5 mm increase in the diameter for either antimicrobial agent, which was tested in combination with clavulanic acid versus its zone tested alone, all are exhibited in muller hinton agar (MHA) after 24 h incubation. The CLSI guidelines values of tested antibiotics was interpreted by chart value, which is effectively presented in Table 1. Further, the Hexa G-minus 24 disc method was validated by using improved ESBL detection Ezy MIC™ E-test stripe, the upper part of the disc of MIX-ceftazidime & cefotaxime (Mixture) - 0.125–16 μg/mL) +/MIX (ceftazidime and lower part of the disc of cetazidime (Mixture) + Clavulanic acid - 0.32–4 μg/mL) calibrated with MIC reading scales in μg/mL (CLSI, 2016) were performed against both the uropathogens. If the ESBLs is positive, the beta-lactamase inhibitor discs with enhanced zone of inhibition was observed, in which the zone of inhibition was $\geq$0.5 and $\geq$1 for CAZ and CTX respectively, and $\geq$8 for cefotaxime/cefotaxime + cefotaxime/clavulanic acid or ceftazidime/ceftazidime + clavulanic acid. In MIC stripe method, for each strip, the point of intersection between the inhibition eclipse and zone of edge in the strips were considered as MIC [14,15].

Molecular screening of ESBLs genes

The highly antibacterial resistant bacteria were chosen, and their TEM. SHV and CTX-m genes were detected by multi plux PCR screening method followed by previously reported evidences of Giovanni et al., [16]. Briefly, high resistant phenotypic bacteria were chosen, and DNA of samples was heated in Distilled water with 30 min. After, the samples were ultracentrifuged at 1000 rpm with 20 min time interval. After successful washing and centrifugation, the samples were added with 150 μg/mL of isopropanol and mixed thoroughly and followed by centrifugation. After, the samples were diluted into TE buffer and continue to run agarose gel electrophoresis for available DNS confirmation. Following, the pellets were collected after remove the supernatant and bacterial DNA containing pellet sample was subjected to PCR amplification using RT-PCR instruments (Shimadzhu, Japan). For detection of all the ESBLs genes, we have used universal primers including ACG CTC ACC GGC TCC AGA TTT AT CGC CGC ATA CAC TAT TCT CAG AAT GA, AGG TGC TCA TCA TGG GAA AG CTT TAT CGG CCC TCA CTC AA and TGG GTR AAR TAR GTS ACC AGA AYC AGC GG ATG TGC AGY ACC AGT AAR GTK ATG GC for TEM (445 bp), SHV (237bp) and CTX-m (593bp). The reactions were arranged for PCR run such as 2x Promega PCR master plus each primer in respective amount and also used 2 μl of template DNA. Each cycle were started with designed parameters such as like initial denaturation with gradual time interval and extension temperature with 95 °C for 10 min and continue to follow 94 °C, 62 °C, and 72 °C for 30 s, 60 s and 90 s time interval respectively. After electrophoresis, the molecular weights of the genes were detected based on the comparison of marker with 100 bP ladder.

Table 1
The Hexa discs zone interpretation chart of CLSI guidelines against *P. aeruginosa* and *K. pnumoniae*.

S No.	Antibiotics	CLSI guidelines for MDRs of Pseudomonadaceae and Enterobacteriaceae (CLSI)			Test pathogens
		S (mm)	I (mm)	R (mm)	
1	Ceftazidime (CAZ-30 μg)	21	18–20	19	*P. aeruginosa* and *K. pnumoniae*
2	Ceftazidime/clavulanic acid (CAC – 30/10 μg)	27	22–26	20	*P. aeruginosa* and *K. pnumoniae*
3	Cefotaxime (CTX-30 μg)	24	21–23	20	*P. aeruginosa* and *K. pnumoniae*
4	Cefotaxime/clavulanic acid (CEC – 30/10 μg)	27	27–29	19	*P. aeruginosa* and *K. pnumoniae*
5	Cefpodoxime (CPD -10 μg)	28	26–28	17	*P. aeruginosa* and *K. pnumoniae*
6	Cefpodoxime/clavulanic acid (CCL – 10/5 μg)	29	28–29	16	*P. aeruginosa* and *K. pnumoniae*
7	Cepime (CPM-30)	26	18–22	23	*P. aeruginosa* and *K. pnumoniae*
8	Ceftriaxone (CTR-30 μg)	23	16–18	17	*P. aeruginosa* and *K. pnumoniae*
9	Amoxclav (AMC)	26	22–24	24	*P. aeruginosa* and *K. pnumoniae*
10	Ciprofloxacin (Cip-5 μg)	21	16–20	15	*P. aeruginosa* and *K. pnumoniae*
11	Co-Trimoxazole (Cot-25 μg)	16	11–15	10	*P. aeruginosa* and *K. pnumoniae*
12	Gentamycin (GEN-10 μg)	13	9–11	10	*P. aeruginosa* and *K. pnumoniae*
13	Imipenom (IMP-10 μg)	26	23–25	22	*P. aeruginosa* and *K. pnumoniae*
14	Ticarcillin/clavulanic acid (TCC-75/10 μg)	24	21–23	21	*P. aeruginosa* and *K. pnumoniae*
15	Meropenem (MRP-10 μg)	26	21–22	22	*P. aeruginosa* and *K. pnumoniae*
16	Ertapenem (ETP-10 μg)	20	17–19	21	*P. aeruginosa* and *K. pnumoniae*
17	Cefoperazone (CFS-75/10 μg)	22	13–16	13	*P. aeruginosa* and *K. pnumoniae*
18	Piperacill/tazobactam (PIT- 100/10 μg)	23	14–17	14	*P. aeruginosa* and *K. pnumoniae*
19	Nalidixic acid (NA-30 μg)	23	18–21	13	*P. aeruginosa* and *K. pnumoniae*
20	Ampicillin (AMP-30 μg)	22	17–19	12	*P. aeruginosa* and *K. pnumoniae*
21	Cephalothin (CEP-30 μg)	22	22–23	16	*P. aeruginosa* and *K. pnumoniae*
22	Ampicillin (AMP-10 μg)	18	14–17	13	*P. aeruginosa* and *K. pnumoniae*
23	Norfloxacin (NF-10 μg)	17	13–16	12	*P. aeruginosa* and *K. pnumoniae*
24	Netillin (NET-10 μg)	16	12–14	10	*P. aeruginosa* and *K. pnumoniae*

Result and discussion

Detection of multi drug resistant profile

Based on the CLSI guideline, the resistant rate was 80% among the 100 strains of selected uropathogens due to the antibiotic susceptibility pattern. In result, almost all the uropathogens were developed resistance to at least one of the UTI panel Hexa disc antibiotics. In contrast, 10% of the strains entirely resistant to all the tested UTI panel of Hexa disc including cephalosporin (CAZ and CTX), carbapenems (IMP and MRP) and fluoroquinolone (Cip and FOS). According to the World health organization report, the pathogens were classified based on the antimicrobial resistant effect and grouped in three priorities; critical, high and medium. Among the priority list, carbapenems resistant GNB was categorized in first ever list of critical, especially carbapenems resistant *P. aeruginosa* and *K. pneumoniae*. Based on the WHO report, we have chosen each one of the *P. aeruginosa* and *K. pneumoniae* for this study due to the highest degree of resistance against all the classes' antibiotics [17]. Based on the CLSI guidelines, the selected bacterial strains were also tested against IPM and MRP of HX012 and it confirmed as cabapenems resistant. The result of both the bacterial strains were more correlated with WHO report, and suggested that the selected uropathogens as carbapenems resistant bacteria. In HX066 and HX033 against *P. aeruginosa*, the IPM, TCC, CPZ and AMP, NX, NA were showed 10, 5 and 4 mm and 10, 14 and 8 mm zone of inhibition were showed respectively (Fig. 1a, b). Whereas, CAZ, MET and AMC, COT of the HX077 and HX012 were showed 16, 22 and 10, 18 mm against *K. pneumoniea* respectively (Fig. 2a, b). Among the Hexa antibiotic disc concentrations, the NIT CEP against *P. aeruginosa*, and CFS, PIT against *K. pnumoniae* were very high (75/10, 30/10, 100/10 and 300 μg) compared with other antibiotics. The antibiotics and their inhibition variations were available in Fig. 3. Similarly, both the bacteria were developed resistant against all the antibiotics, also in high concentration due to the production of efflux pump, ESBL production, biofilm formation and enzyme production [13]. Here, the minimum zones of antibiotics against both the uropathogens were indicated that the antibiotics as more sensitive due to the resilient nature of the bacteria [18]. The exhibited zones could not perform better against cell wall damage and target modification in selected uropathogens (CLSI). Hence, our result confirmed that the selected uropathogens were MDR strains as well as carbapenoms resistant strains. The CLSI guidelines interpretation chart of Hexa discs containing antibiotics zone against tested uropathogens were summarized in Table 1. Our result was most accordance with earliest result of Rajivgandhi et al. [19]. the MDR effect of GNB was more in UTIs due to the over

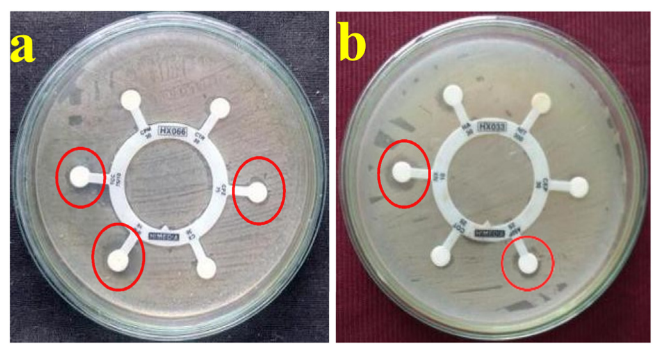

Fig. 1. Detection of multi drug resistant *P. aeruginosa* by Hexa disc diffusion method using HX066 (a), HX033 (b).

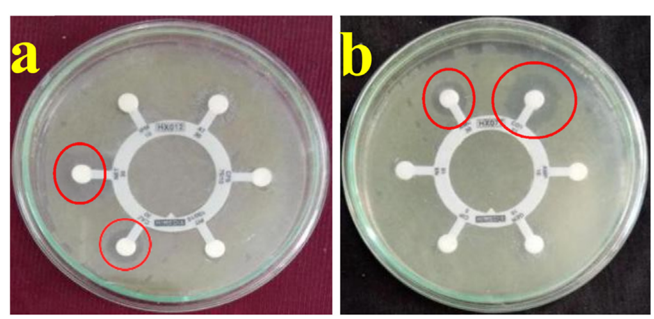

Fig. 2. Detection of multi drug resistant *K. Pneumoniae* by Hexa disc diffusion method using Hexa discs of HX077 (a), HX012 (b).

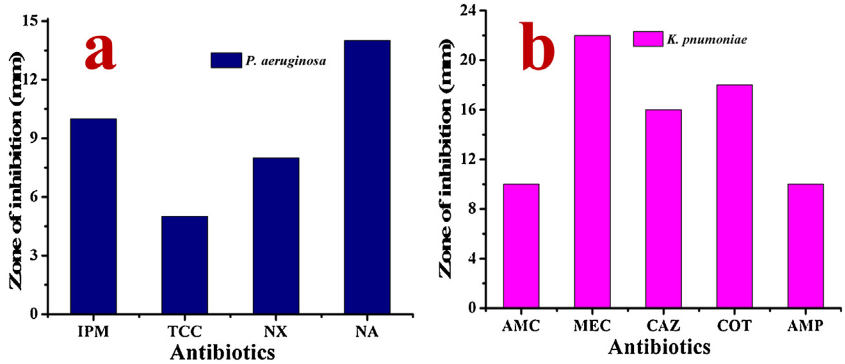

Fig. 3. Partial inhibition zones Differentiation of using respective antibiotics against *P.aeruginosa* (a) and *K. pneumonia* (b).

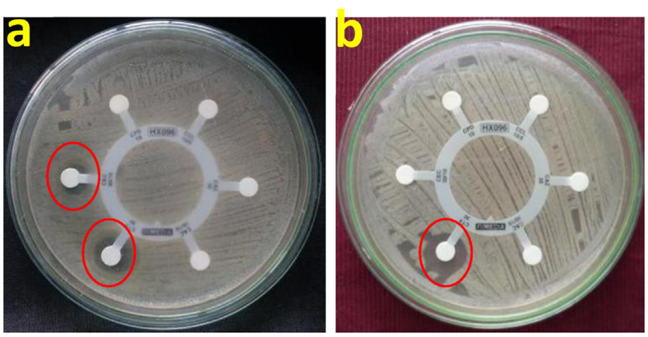

Fig. 4. Screening of ESBL production against *P. aeruginosa* (a) and *K. pnumoniae* (b) by Hexa G-minus 24-disc.

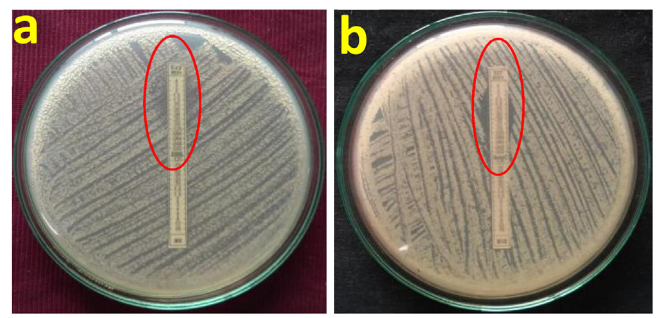

Fig. 5. Confirmation of ESBL production in *P. aeruginosa* (a) and *K. pnumoniae* (b) by MIC stripe methods.

use and extensive use of antibiotics. The partial inhibition zone of ciprofloxacin and third generation chaphalosporin antibiotics were more sensitive to Enterobacteriaceae due to the production of ESBLs [20].

Detection of ESBL production

The ESBL producing ability of both the MDR effect of *P. aeruginosa* and *K. pnumoniae* were confirmed by Hexa G-minus 24-disc diffusion method. After 24 h incubation, the CTX of both the discs were exhibited 18 and 16 mm zone of inhibition against both the uropathogens and CAC against *P. aeruginosa* were showed 12 mm zone respectively. No any zones of inhibition around the rest of other antibiotics were also observed (Fig. 4a, b). The ESBL identification discs of CAZ, CTX and CPD were showed $\geq$22 mm, $\geq$27 and $\geq$24 zones of clearance against both the uropathogens, and this result indicated that the selected uropathogen were ESBL producer (Fig. 6a). Similarly, combination discs of the CAC, CEC exhibited $\geq$5 mm zone of clearance against both the uropathogens were observed, and the exhibited result was strongly proved both the MDRs effect was acquired by ESBL production. Also, the exhibited zones were showed within the resistant level of antibiotics and were able to degrade the β-lactum ring in selected antibiotics due to the oxyimino side chain cleavage [21]. The result of ESBLs identification of combination discs methods were proved, selected uropathogens were able to produce ESBLs.

Further, the exhibited result of 0.35, 0.39 mm of the MIX discs against both uropathogens were confirmed that the selected strains were ESBL producers (Fig. 5a, b). Whereas, the no zone of clearance was observed at lower end of the both discs against respective pathogens were also observed. In the MIX end, the MIC range of 0.32–6 μg of the respective antibiotics and lower end of 1.25–16 μg of the respective antibiotics were used for detection of ESBL production. The result of MIC range was exhibited at $\geq$8, which clearly evidenced that the selected uropathogens were ESBL producer (Fig. 6b). The similar study was conducted by Parveez ahamed et al., 2017 [22] and reported that the MIC stripe was confirmation method for ESBL producing pathogens. The results obtained in the present study agree with earlier findings reported by Stuart et al. [23], and MIC stripe method was successful method for detection of ESBL producing bacteria. Hence, both the phenotypic ESBL identification methods were confirmed that the selected uropathogens were developed MDR effect due to the ESBL production and it may cause severe infections in human urinary tract including blockage of urinary bladder, kidney stones and bacteriuria [24]. ESBLs are co-resistance to multiple other classes of antibiotics, resulting in limitation of therapeutic option, this will be exceeding the mortality rate of the other life-threatening disease [25]. In recent decades, the emergence of β-lactamase inhibitors results in failure of interactive antibiotic treatment. Discovery of novel and potential synergistic antimicrobial agents to overcome the bacterial resistance are urgently required [26].

Molecular gene identification

In multi plux PCR experiment for detection of ESBLs gene is important evident to confirm the bacterial enzymatic activity. Also, this method was used to confirm, whether the bacteria could fulfill the phenotypic identification was correct or not [18]. Based on this approach, the current result was more supported to phenotypic results and confirmed that the pathogens were able to develop resistance against any antibiotics with high virulence. In this genetic detection, the result was confirmed that the TEM, SHV and CTX-m were present in the both the *P. aeruginosa* and *K. pneumoniae* culture. In 2017, WHO reported that the ESBLs producing bacteria were comes under Class I category and it cause severe infection in human. The class 1 pathogens are very effective in treatment due to the modification of their structure. If they are changing the virulence factors more and multiple virulence factors are not

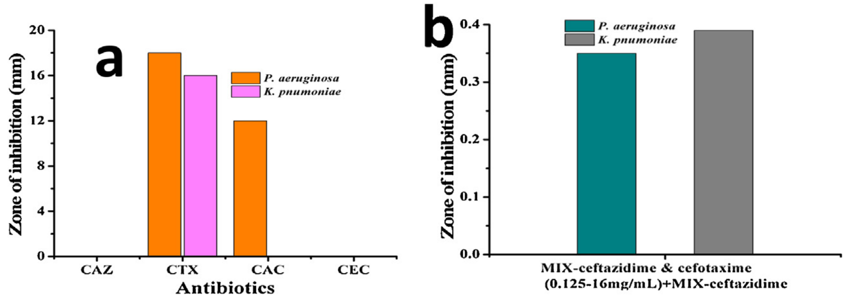

Fig. 6. All the tested Hexa G-minus 24-disc antibiotics of were developed resistance against both the uropathogens (a) and the partial inhibition zone variation of MIC stripe against both the uropathogens (b).

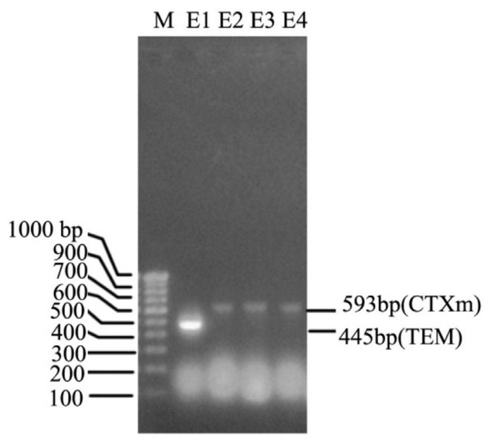

Fig. 7. Multiplux PCR analysis of ESBL genes detection in *P. aeruginosa* (MN310553) and *K. pnumoniae* (MN368594) (b).

susceptible to any antibiotics [27]. This problem is persisting till end of the human period due to the continuous cause of similar infections like urinary tract, urethritis, kidney damage, kidney stone and etc [28]. Current result was more agreed by WHO reports and ESBLs producing *K. pneumoniae* and *P. aeruginosa* are most dangerous pathogens and prevention also more difficult due to the multiple virulence factors development such as exopolysachharides, quorum sensing, biofilm formation and etc [29]. In addition, most of the urinary tract pathogens are ESBLs producers and cause dangerous infections. In our result, the genes of TEM, SHV and CTX-m were shown in the position of 445, 237 and 593 bp respectively (Fig. 7). The similar results were reported previously by Ahmad et al. [30], and most of the urinary pathogens are ESBLs producers and the treatment also very difficult. Recently, Ziheng et al. [31] reported that the Class 1 *K. pneumonia* and *P. aeruginosa* were very dangerous bacteria and cause more in-depth region of kidney. Previously, Balamuralikrishnan et al. [32], reported that the multiplex PCR method is very effective method to detection of ESBL genes in multi drug resistant bacteria, and it also helped to confirm the phenotypic disc diffusion methods.

Conclusion

In this study, *P. aeruginosa* and *K. pneumoniae* were identified as multi drug resistant effect and it was more susceptible to current antibiotics. Importantly, the specific Hexa discs were tested against selected uropathogens and they were indicated 10, 5 and 4 mm and 10, 14 and 8 mm zone of inhibition for the antibiotics of IPM, TCC, CPZ and AMP, NX, NA. They were confirmed that the pathogen were multi drug resistant pathogens. Consequently, the ESBLs and carbapenemase production was very highly present in both the pathogens and were confirmed by phenotypic confirmation assays. Further, the molecular identification of both the bacteria was shown TEM. SHV and CTX-m genes. Altogethr, the present result was concluded that the selected uropathogens were ESBLs producers and it treatment options were very less.

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Competing interests

None declared.

Ethical approval

The samples were approved by the ethics review committee (S. No of IEC Management office: DM/2016/101/55) from the Department of Microbiology, Bharathidasan University, Tiruchirappalli, Tamil Nadu, India. The Permission was sought from the hospital and laboratory authorities. The ethical principles of scientific research as well as related national laws and regulations were adhered to.

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